

Foundations of Risk Management 目录

题号 7 考察: Credit Risk [■□□□□]	8
题号 24 考察: Finance: Fixed Income [■□□□□]	19
题号 36 考察: Quantitative Analysis: Copulas [■□□□□]	26

Quantitative Analysis 目录

题号 1 考察: Market Risk [■ ■ □]	5
题号 3 考察: Operational Risk and Resiliency [■ ■ □]	6
题号 4 考察: Market Liquidity [■ ■ □]	7
题号 5 考察: Finance: Portfolio Management [■ ■ □]	7
题号 6 考察: Derivatives: Options [■ ■ □]	8
题号 8 考察: Finance: Regulatory Capital [■ ■ □]	9
题号 9 考察: Market Risk: Liquidity Risk [■ ■ □]	10
题号 10 考察: Portfolio Management: Fundamental Law of Active Management [■ ■ □]	10
题号 11 考察: Term Structure Models [■ ■ □]	10
题号 12 考察: Finance: Credit Risk [■ ■ □]	12
题号 13 考察: Finance: Basel Accords [■ ■ □]	12
题号 14 考察: Financial Markets and Products [■ ■ □]	13
题号 15 考察: Finance: CAPM [■ ■ □]	13
题号 16 考察: Fixed Income: Vasicek Model [■ ■ □]	14
题号 17 考察: Finance: Financial Statement Analysis [■ ■ □]	15
题号 18 考察: Operational Risk [■ ■ □]	15
题号 19 考察: Liquidity Risk Management [■ ■ □]	16
题号 20 考察: Finance: Factor Theory [■ ■ □]	17
题号 21 考察: Fixed Income [■ ■ □]	17
题号 22 考察: Credit Risk Management [■ ■ □]	18
题号 23 考察: Market Risk [■ ■ □]	19
题号 26 考察: Fixed Income Analysis [■ ■ □]	20
题号 27 考察: Credit Risk	21
题号 28 考察: Operational Risk [■ ■ □]	22
题号 29 考察: Finance [■ ■ □]	22
题号 30 考察: Finance: Risk Management [■ ■ □]	23
题号 31 考察: Fixed Income: Hedging [■ ■ □]	24
题号 32 考察: Credit Risk: Wrong-Way Risk [■ ■ □]	24
题号 33 考察: Operational Risk: Model Risk Management [■ ■ □]	25
题号 34 考察: Market Risk: Interest Rate Risk [■ ■ □]	25
题号 37 考察: Credit Risk: Retail vs Corporate [■ ■ □]	26
题号 38 考察: Operational Risk [■ ■ □]	27
题号 40 考察: Finance: Hedge Funds [■ ■ □]	29

Financial Markets and Products 目录

题号 2 考察: Credit Risk [■ ■ □]	5
题号 25 考察: Factor Investing [■ ■ □]	19

题号 35 考察: Finance: Due Diligence 26

1. Consider an exposure with a value of \$30.0 million. In L(+)/P(-) format, its expected loss is 4.0% with volatility of 15.0%; i.e., $\mu(L/P) = +4.0\%$, $\sigma = 15.0\%$. Conveniently, the exposure happens to have a normal distribution. Which is **nearest** to the single-period 97.5% expected shortfall (ES)?

- A. 6.60 million
- B. 9.38 million
- C. 11.72 million
- D. 14.50 million

答案:C

解析:

C 选项正确。 If the distribution is normal, the 97.5% ES is approximated by the 99.0% VaR. In this case, we can use $\$30.0 * (4.0\% + 15\% * 2.33) = \$30.0 * 38.950\% = \$11.69$. The exact 97.5% ES is given by \$11.7210, so that is given as the correct choice.

考察: Market Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

2. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability

答案:D

解析:

D 选项正确。 Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability.

Wrong-way (credit) risk is adverse correlation between the exposure and the credit quality, so wrong-way risk is a positive (negative) correlation between exposure and default probability (credit quality). This is wrong-way risk because greater exposure is associated with a lower likelihood of payment. Algebraically, wrong-way (or right-way) risk is some dependence between exposure/EAD and PD. We normally write $EL\% = PD \times EAD$ but that assumes independence between PD and EAD, so a way to think about WWR is simply when $EL > PD \times EAD$, because $EL \neq PD \times EAD$ is also a condition for non-independence. In regard to (A), (B) and (C), each is an example of RIGHT-WAY risk.

考察: Credit Risk 重要程度: [■ ■ □]

科目: Financial Markets and Products

3. In December 2017, the Basel Committee (BCBS) made several revisions to the Basel III framework. This included revisions to both the standardized approach to credit risk (“Credit risk accounts for the bulk of most banks’ risk-taking activities and hence their regulatory capital requirements. The standardized approach is used by the majority of banks around the world, including in non-Basel Committee jurisdictions,” wrote the Committee) and the internal ratings-based (IRB) approaches to credit risk (“The financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes,” wrote the Committee).¹

In regard to these December 2017 revisions to credit risk approaches in Basel III, which of the following statements is TRUE?

¹ “High-level summary of Basel III reforms,” (Basel Committee on Banking Supervision Publication, December 2017)

- A. In the standardized approach (SA) for residential real estate exposures, the risk weights are lower if the loan-to-value (LTV) falls into a lower band
- B. The standardized approach (SA) eliminated the treatment of unrated exposures because all exposures are required to have an external credit rating
- C. The internal ratings-based (IRB) approach removed the option to use the foundation IRB (F-IRB) because the advanced IRB (A-IRB) must be used for all asset classes
- D. The internal ratings-based (IRB) approach removed the parameter input floors (in the case of PD, LGD, and EAD) to increase RWA variability across different banks

答案:A

解析:

Option A is correct. In the standardized approach (SA) for residential real estate exposures, the risk weights is lower if the loan-to-value (LTV) falls into a lower band.

See below. Writes BCBS (High-level summary of Basel III reforms, December 2017): “For residential real estate exposures, more risk-sensitive approaches have been developed, whereby risk weights vary based on the LTV ratio of the mortgage (instead of the existing single risk weight) and in ways that better reflect differences in market structures.”

Residential real estate exposures								
LTV bands	Below 50%	50% to 60%	60% to 70%	70% to 80%	80% to 90%	90% to 100%	above 100%	Criteria not met
General RRE								
Whole loan approach RW	20%	25%	30%	40%	50%	70%	RW of counterparty	
Loan-splitting approach ² RW	20%	RW of counterparty						RW of counterparty
Income-producing residential real estate (IPRRE)								
Whole loan approach RW	30%	35%	45%	60%	75%	105%	150%	

In regard to (B), (C) and (D), each is FALSE. Instead, the following are true:

- The Basel III reforms included a more granular approach to unrated exposures and reduced the “mechanistic reliance on credit ratings”
- In regard to the internal ratings-based approaches, GARP writes, “the financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes. To address these shortcomings, the Committee has made the following revisions to the IRB approaches: (i) removed the option to use the advanced IRB (A-IRB) approach for certain asset classes; (ii) adopted “input” floors (for metrics such as probabilities of default (PD) and loss-given-default (LGD)) to ensure a minimum level of conservatism in model parameters for asset classes where the IRB approaches remain available; and (iii) provided greater specification of parameter estimation practices to reduce RWA variability.” — Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.
- In regard to the Specification of Input Floors: “The revised IRB framework also introduces minimum floor values for bank-estimated IRB parameters that are used as inputs to the calculation of RWA. These include PD floors for both the F-IRB and A-IRB approaches, and LGD and EAD floors for the A-IRB approach. In some cases, these floors consist of recalibrated values of the existing Basel II floors. In other cases, the floors represent new constraints for banks’ IRB models.” — Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.

考察: Operational Risk and Resiliency 重要程度: ■■■□

科目: Quantitative Analysis

4. An asset that is not very liquid is quoted bid \$89.00, offer \$96.00. Which is **nearest** to the proportional bid-offer spread?

A. 0.0378

B. 0.0757

C. 0.1538

D. 7.000

答案:B

解析:

B 选项正确。 0.0757 . Because $(96.00 - 89.00)/92.50 = 0.075676$

According to Hull (24.1.2. Measuring Market Liquidity), the two measures of bid-offer spread are:

- Dollar bid-offer spread = $p = \text{Offer price} - \text{Bid price}$
- Proportional bid-offer spread = $s = (\text{Offer price} - \text{Bid price}) / \text{Mid-market price}$

(Source: John C. Hull, Risk Management and Financial Institutions, 5th Edition (Hoboken, NJ: John Wiley & Sons, 2018))

考察: Market Liquidity 重要程度: [■ ■ □]

科目: Quantitative Analysis

5. Richard conducted a factor regression for his firm’s Flagship Portfolio. He used the three-factor Fama-French specification where the factors are MKT (the market), SMB (small stocks minus big stocks), and HML (value stocks minus growth stocks). His factor regression results are displayed below.

<i>Flagship excess returns, $R(p) - R_f$, Fama-French benchmark (monthly)</i>		
	Coeffi- cient	Standard Error (SE)
Alpha, α	0.300%	0.033%
MKT Loading	1.250	0.096
SMB Loading	0.640	0.043
HML Loading	-0.920	0.066
Adjusted R^2	0.290	

Please note the regression is based on monthly returns. Each of the following statements is TRUE about these factor regression results EXCEPT which is false?

- A. The portfolio’s style appears to growth-oriented, small-cap
- B. This portfolio manager(s) appear to be skillful; that is, the alpha is significant
- C. The low adjusted R^2 and low standard errors imply these factors are not appropriate to this portfolio
- D. The implied CAPM benchmark is leveraged as it borrows cash (i.e., T-bills) to invest more than 100% (i.e., $> \$1.0$) in the market portfolio

答案:C

解析:

C 选项正确。 C. FALSE: The factors are each significant and the adjusted R^2 is relatively high for empirical finance. In regard to (A), (B) and (D), each is true.

考察: Finance: Portfolio Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

6. The riskfree rate is 5.0% while the current stock price is \$80.00. Two most liquid options both have a strike price of \$70.00 such that the ratio $K/S(0) = 0.90$. In addition to the same strike price, both options have the same maturity of three months, $T = 0.25$ years. The implied volatility of the call is 30.0%, and the implied volatility of the put is 38.0%. What is the trade that exploits this scenario, if any?
- A. Buy call, sell put, and short the stock
 - B. Buy put, sell call, and buy the stock
 - C. Buy the call, buy the put, and short a straddle
 - D. Sell the call, sell the put, and buy a straddle

答案:A

解析:

A 选项正确。 True: Buy call, sell put, and short the stock

考察: Derivatives: Options 重要程度: [■■■□]

科目: Quantitative Analysis

7. Consider a netting set with two derivative positions: an interest rate swap (IRS) with a marginal CVA of -1.450 and a cross-currency swap (CCS) with a marginal CVA of -1.550 . If the first transaction is the cross-currency swap (CCS) and its incremental CVA is -2.100 , then what is the incremental CVA of the IRS?
- A. -0.900
 - B. -1.100
 - C. -1.550
 - D. -2.220

答案:A

解析:

A 选项正确。 The total CVA is the sum of marginal CVAs and therefore must be $-1.450 + (-1.550) = -3.000$. The incremental CVAs must sum to the total (netted) CVA such that, if the IRS is second in sequence, its incremental CVA is equal to $-3.000 - (-2.100) = -0.900$.

考察: Credit Risk 重要程度: [■□□□□]

科目: Foundations of Risk Management

8. The central feature of Basel III's first pillar is the regulatory capital (i.e., CET1, Tier 1, and total capital) requirements which require minimum fractions of risk-weighted assets (RWA). In addition to these minimums, as of early 2019 (when the buffers were fully phased in), Basel specifies three buffers: the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB). This is all very confusing, so you ask your colleague Mary to summarize these additional buffers and their motivations. She makes the following five points:
- I. For all three, the additional buffer must be CET1
 - II. For all three, breach of the buffer requirements implies the bank's ability to pay dividends will be restricted
 - III. The CCB requires an additional 2.5% of CET1 capital and is meant to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred.
 - IV. The additional G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} https://www.bis.org/fsi/fsisummary_sib_framework.htm and is meant to reduce the likelihood and severity of the failure of a global systemically important financial institution
 - V. The CCyB requirement varies between zero and 2.5% and is meant to protect the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risks

Is Mary correct in her summary?

- A. No, unfortunately, none of her statements is correct
- B. I. and II. are correct, but the others (III., IV., and V.) are wrong
- C. III., IV., and V. are correct, but the others (I. and II.) are wrong
- D. Yes, all five statements are correct

答案:D

解析:

D 选项正确。 Yes, all five statements are correct

- In regard to true statements (I.) and (II.), the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB) each require common equity Tier 1 (CET1) which is a buffer of the highest quality. Further, breach of any of them implies restrictions on dividend payments (in addition to other supervisor tools such as restriction on executive bonuses)
- In regard to true statement (III.), the BCBS says that “the capital conservation buffer [CCB] was introduced to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred” (see https://www.bis.org/fsi/fsisummaries/b3_capital.htm) and GARP says “In the case of the CCB, the rationale roughly follows that for the Prompt Corrective Action (PCA) system built into U.S. capital regulation beginning in 199; i.e., a bank with ratios that begin to approach the minimums should be subject to increasingly stringent supervisory intervention in order to induce a return to well-capitalized status).” -2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.
- In regard to true statement (IV.), the G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} and, explains GARP, “In the case of the G-SIB buffer, the rationale is similar to that for the CCB but also recognizes the very large costs to society of distress at G-SIBs (and the higher volatility of losses at some of them). Thus, larger buffers are specified to further reduce the chance of failure.”
- In regard to true statement (V.), “The CCyB [countercyclical capital buffer] has two rationales. One is to provide an instrument for macroprudential restraint of overheating; the other is attentive to the cost of capital,” says GARP. Further, explains BCBS, “The countercyclical capital buffer [CCyB] aims to ensure that banking sector capital requirements take account of the macro-financial environment in which banks operate. Its primary objective is to use a buffer of capital to achieve the broader macroprudential goal of protecting the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risk. Due to its countercyclical nature, the countercyclical capital buffer regime may also help to lean against the build-up phase of the credit cycle in the first place. In downturns, the regime should help to reduce the risk that the supply of credit will be constrained by regulatory capital requirements that could undermine the performance of the real economy and result in additional credit losses in the banking system.”
See <https://www.bis.org/bcbs/ccyb/>

Keep in mind the basic requirements of these three additional buffers: the CCB requires +2.5% CET1; the G-SIB requires either {+1.0%, +1.5%, +2.0%, +2.5%, or +3.5%} depending on the bank’s score (see <https://trtl.bz/fsb-g-sibs>); and the CCyB requires between zero and 2.5%.

考察: Finance: Regulatory Capital 重要程度: [■ ■ ■ □]

科目: Quantitative Analysis

9. A portfolio holds 100,000 shares of a stock and this single position has a value of \$3.0 million. The stock is quoted bid \$29.00, offer \$31.00. The stock’s daily volatility is 1.43% or 143 basis points. For purposes of value at risk (VaR), we will assume the stock’s arithmetic returns are normally distributed (aka, normal VaR) and the expected daily return rounds to zero (under these assumptions absolute VaR is identical to relative VaR). Which is **NEAREST** to the position’s one-day 99.0% confident liquidity-adjusted value at risk (LVar)?

- A. \$15,000
- B. \$80,000

- C. \$200,000
- D. \$1.0 million

答案:C
解析:

Option C is correct. The one-day 99.0% VaR is given by $-0 + 1.43\% * 2.326 * \$3.0 \text{ million} = \$99,800$.
The liquidity cost (LC) is given by $(31.00 - 29.00)/30.00 * 0.5 * \$3.0 \text{ million} = \$100,000$.
The one-day 99.0% VaR is therefore $99,800 + 100,000 = 199,800$

考察: Market Risk: Liquidity Risk 重要程度: [■■■□]
科目: Quantitative Analysis

- 10.** According to Grinold, typical or illustrative information ratios (IR) include: 0.500 is good, 0.750 is very good and 1.00 is exceptional. In regard to risk aversion, λ , 0.150 is restrained, 0.100 is moderate, and 0.050 is aggressive. Patricia is a manager with a “very good” information ratio of 0.750 and a “moderate” risk aversion of 0.10. If she shifts from this moderate risk aversion to an aggressive risk aversion of 0.050, what is the impact on the optimal level of active risk?
- A. No effect
 - B. Optimal active risk reduces by one-half
 - C. Optimal active risk Increases by about +41%
 - D. Optimal active risk doubles

答案:D
解析:

D 选项正确。 True: Optimal active risk doubles As the optimal level of active risk is given by $IR/(2 * \lambda)$, Patricia doubles her optimal level when she goes from $0.750/(2 * 0.10) = 3.750$ to $0.750/(2 * 0.050) = 7.500$
考察: Portfolio Management: Fundamental Law of Active Management 重要程度: [■■■□]
科目: Quantitative Analysis

- 11.** For his term structure model, Peter considered the Vasicek model, but on reflection, he prefers the Cox-Ingersoll-Ross (CIR). In regard to the CIR, which of the following statements is TRUE?
- A. A drawback of the CIR is that it does not mean revert to a long-term rate
 - B. In the CIR, the drift is time-dependent such that CIR is a no-arbitrage model
 - C. An advantage of the CIR is that the terminal distribution of the short rate is eventually (e.g., > 10 years) normally distributed
 - D. Unlike the Vasicek model’s constant volatility, the CIR’s basis point volatility is not constant, and this ensures the short rate cannot become negative

答案:D
解析:

D 选项正确。 True: Unlike the Vasicek model’s constant volatility, the CIR’s basis point volatility is not constant, and this ensures the short rate cannot become negative.
In regard to (A), (B), and (C), each is FALSE.
考察: Term Structure Models 重要程度: [■■■□]
科目: Quantitative Analysis

- 12.** Jason is a small business owner specializing in the wholesale of steel. Jason worked hard on his business, so he purchased insurance to help safeguard his company on January 4th, 2023. Later that month, on January 31st, he made a significant

payment to secure a shipment of steel that will arrive on February 2nd, 2023. Jason delivered the steel to two clients on February 8th and 9th; however, he did not receive payment from the clients until February 13th. After receiving the proceeds Jason deposited the money at ABC bank.

Among these transaction dates, on which did Jason increase his credit risk?

- A. None of the transactions increase his credit risk
- B. His credit risk increased on the Jan dates (Jan 4th and 31st) but was unchanged on Feb dates (Feb 8th, 9th and 13th)
- C. His credit risk increased on the Feb dates (Feb 8th, 9th, and 13th) but was unchanged on Jan dates (Jan 4th and 31st)
- D. His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)

答案:D

解析:

D 选项正确。 True: His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)

- Jan 4th: Purchasing insurance increases credit risk because Jason is buying a policy that will not be delivered if ever once an event occurs. *Contingent claims* are a type of credit risk. According to the text, “A sixth type of transaction is a special case of a claim on an asset—a contingent claim. The claim is contingent on certain events occurring, such as a loss covered by an insurance policy. At policy inception, the policyholder has no claim on the insurer. However, once the insured suffers a covered loss, the insured has a claim. If the insurer fails to pay the claim, this would constitute a credit loss.” (†)
- Jan 31st: He prepays for an order and thus has the possibility of his counterpart not delivering on their obligation. According to the text, “*Prepayment of goods and services* is a fourth type of transaction that involves credit risk. Delivery is expected at a certain time and of a certain quality and/or performance, and the failure of the counterparty may lead to the loss of the advanced payments and also generates business interruption costs.” (†)
- February 8th and 9th: Jason delivered an order of steel to his clients. Either of his clients could have not delivered on their obligation to pay him, and thus, Jason increased his credit risk. According to the text, “The third type is the sale of a product or a service without immediate cash payment. The seller sends an invoice to the buyer after the product has been shipped or the service performed, and the buyer has a few weeks to pay. This is known as an *account receivable*.” (†)
- February 13th: He deposited the proceeds of his sale and his earnings from last quarter at ABC Bank. Jason granted a third-party custody over his assets, and thus, he has the risk of the bank not being able to repay his assets. Per the text, “A fifth type of transaction that creates credit risk involves a party’ s claim on an asset in the custody of or under the management of another party, such as a bank deposit.” (†)

Credit risk can be generated in myriad ways. The text(†) lists seven:

1. Lending: Risk of non-repayment from the borrower to the lender.
2. Leases: Risk that the lessee will fail to make future payments for the use of an asset.
3. Accounts Receivable (Sale of Product/Service without Immediate Payment): Risk that the buyer will not pay after receiving the product or service.
4. Prepayment of Goods and Services: Risk that the counterparty will not deliver or meet the quality/performance after receiving payment.
5. Deposits and Bank Balances: Risk that the financial institution holding deposits or managing assets will default.
6. Contingent Claims (e.g., Insurance, Pension Fund Claims): Risk that the insurer or sponsor will not be able to pay out the claim or pension liabilities.
7. Derivative Exposures: Risk from future payment obligations in derivative transactions like swaps or futures, which are dependent on the underlying asset’ s market value.

(†) Sylvain Bouteille and Diane Coogan-Pushner, *The Handbook of Credit Risk Management: Originating, Assessing, and Managing Credit Exposures* (2nd Edition, Hoboken, NJ: John Wiley & Sons, 2022).

考察: Finance: Credit Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

13. Peter is studying for the Financial Risk Manager (FRM) exam, and he thinks the historical evolution of the Basel regulations is confusing. In particular, he wants to better understand the difference between the original Basel I accord (aka, the 1988 Basel Accord that was implemented by 1992) and Basel II, which was finally published over ten years later in 2004. His colleague Mary explains that, in comparison to the original accord, Basel II contained four significant innovations. Specifically, in comparison to the original Basel I accord (which do include the 1995/1996 Amendments) that came before, Mary says that significant innovations in Basel II included the following:

- I. In addition to credit risk and market risk, Basel II required capital for operational risk
- II. Basel II eliminated risk weights and risk-weighted assets (RWA) and replaced them with direct calculation of risk charges
- III. Basel II contained specific requirements for supervision related to capital and risk management (Pillar 2) and required public disclosures (Pillar 3)
- IV. To fine-tune the accord’ s design, Basel II made repeated use of Quantitative Impact Studies (QIS) to which banks contributed data (i.e., feedback) that was analyzed by supervisors

Is Mary correct?

- A. Yes, all four statements are correct
- B. I. and II. are correct but III and IV are inaccurate
- C. Only II. is inaccurate but I, III, and IV are correct
- D. Unfortunately, none of her statements are correct

答案:C

解析:

Option C is correct. Basel II did NOT eliminate risk-weighted assets (or risk weights); risk-weighted assets (RWA) do continue in Basel III/IV. According to GARP’ s Mark Carey, the four significant innovations of Basel II (as an improvement over the original Basel I accord) were: more sophisticated risk weight formulas; the addition of an operational risk capital charge; Pillars 2 and 3; and the repeated use of the Quantitative Impact Studies (QIS). Many experts consider the additional Pillars to be a highly significant development. As Carey explains, “The three pillars represented a push toward convergence of national practices [especially in the context of the “rules versus principles” disparity]. Specifically, Pillar 2 mandated that supervisors require banks to have more than the minimum amount of capital as well as internal capital adequacy and assessment processes (ICAAP) that take their risk profile into account ... National discretion regarding enforcement of the accord’ s provisions was reduced, and national regulators were to be transparent about their implementation efforts, including those concerning the requirements in excess of the minimums. Pillar 3 required more qualitative and quantitative disclosures, in the hope that pressure from market participants would help improve banks’ practices [aka, market discipline].” (Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition. Pearson Learning Solutions)

考察: Finance: Basel Accords 重要程度: [■ ■ □]

科目: Quantitative Analysis

14. Assume the spot foreign exchange (FX) rate between the US dollar and the Japanese Yen is 110.00 which can be represented as USDJPY 110.00 or JPY 110.00 where USD is the base currency and JPY is the quote currency; aka, quote 110.00 yen per one dollar of the base or “per unit” currency. The USD interest rate is 2.50% and the JPY interest rate is 0.450%. The period is one year, and the compound frequency is annual. If the covered interest rate (CIP) arbitrage framework enforces a perfectly accurate one-year forward exchange rate, then what is the implied one-year “swap rate” which is here simply defined as the difference between the forward FX rate and the spot FX rate; put simply, what is the difference, $F(\text{USDJPY}) - S(\text{USDJPY})$, or $F(\text{USDJPY}) - 110.00$?

A. -2.20 JPY (or -2.20 USDJPY)

B. Zero

C. + 5.80 JPY

D. + 13.70 JPY

答案:A

解析:

A 选项正确。 True: $F - S = -2.20$ USDJPY

If CIP is enforced, then $F - S = S \times [(1 + \text{Rf_JPY})/(1 + \text{Rf_USD}) - 1] = 110.00 \times [(1 + 0.45\%)/(1 + 2.50\%) - 1] = -2.20$.

考察: Financial Markets and Products 重要程度: [■■□]

科目: Quantitative Analysis

15. Volatility is an important risk factor according to Andrew Ang. The strong negative relationship between VIX changes and stock returns is explained by (at least) two dynamics. The first is called the leverage effect. The “second channel is a time-varying risk premium story and is the one the basic CAPM [capital asset pricing model] advocates,” explains Ang.(†) Assume the riskfree rate is 3.0% while the equity risk premium (EPR; aka, the market’s expected excess return) is 6.0% because the market’s expected return is 9.0%. According to the CAPM, and consistent with Ang’s theory on the transmission channel between volatility and stock returns, which of the following should cause a DECLINE in a stock’s price?

A. The VIX Index declines from 21.00 to 19.00

B. The market’s volatility decreases and the stock’s volatility decreases along with it

C. The stock’s volatility decreases, which corresponds to an increase in its leverage via the leverage effect

D. The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged

答案:D

解析:

Option D is correct. The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged.

An increase in the stock’s volatility and/or increase in its correlation to the market, while the market’s volatility is unchanged (aka, ceteris paribus), implies an increase in the stock’s beta. This increases the stock’s expected return. Consequently, the stock’s price should decline.

Ang (emphasis ours): “The correlation between VIX changes and stock returns is -39% , so stocks do badly when volatility is rising. The negative relation between volatility and returns is called the leverage effect. When stock returns drop, the financial leverage of firms increases since debt is approximately constant while the market value of equity has fallen. This makes equities riskier and increases their volatilities. There is another channel where high volatilities lead to low stock returns: **an increase in volatility raises the required return on equity demanded by investors, also leading to a decline in stock prices.** This second channel is a time-varying risk premium story and is the one that the basic CAPM advocates: as market volatility increases, discount rates increase and stock prices must decline today so that future stock returns can be high.” (†)

In regard to (A), (B), and (C), each is false.

(†) Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York: Oxford University Press, 2014).

考察: Finance: CAPM 重要程度: [■■□]

科目: Quantitative Analysis

16. An analyst fits a Vasicek model when the current short-term rate is 4.60%. If the theta, θ , parameter is 13.50%, then each of the following statements is true EXCEPT which is false?

A. A key advantage of the Vasicek model is that it is a relatively simple model

- B. If the half-life is nine years, then we might expect the short rate to reach $\sim 9.0\%$ in nine years or so
- C. The standard deviation of the short rate in ten years is less than it would be without mean reversion
- D. Because it is a multi-factor model, the Vasicek is popularly applied to hedge portfolios that contain short-, medium- and long-term bonds

答案:D

解析:

D 选项正确。 False. Vasicek is a single-factor model; to hedge short-, medium- and long-term bonds would require, for example, the three-factor Gauss+ model.

In regard to (A), (B), and (C), each is TRUE.

考察: Fixed Income: Vasicek Model 重要程度: [■■■]

科目: Quantitative Analysis

17. Firm A, Firm B, and Firm C are three companies in the bottled water industry. All companies have a mix of debt and equity in their capital structure. In 2022 all three companies had credit losses that lowered EBIT. Below is a summary of their financial information for the year 2022.

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for Loan Loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%

Given the above information, which firm has the higher return on equity (ROE)?

- A. Firm C has the highest ROE due to its lower equity position.
- B. Firm A had the higher ROE because of its large debt position.
- C. Without credit losses, firm A would have had the highest ROE.
- D. Without credit losses, firm B would have had the highest ROE.

答案:A

解析:

A 选项正确。 True: Firm C has the highest ROE due to its lower equity position.

- Firm A has a negative ROE (-0.33%) because of its large debt position (as illustrated below). Even after sustaining credit losses the firm still must repay its debt holders. The A sustained heavy credit losses that lowered EBIT. However, even after adjusting for its credit losses Firm A still would have had a smaller ROE than firms B and C.
- Firm B had a positive ROE, but the ROE is very small because of its large equity compared to firms C and A. Firm A sustained a large credit loss that lowered EBIT. However, even after adjusting for its credit losses the firm would have still had a smaller ROE than firm C.
- Firm C had the highest ROE because of its small equity and modest credit loss.

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for loan loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%
Solution			
EBIT	\$75,000	\$75,000	\$75,000
Interest	\$85,000.00	\$34,000.00	\$34,000.00
Tax	0	\$8,610.00	\$8,610.00
NI	(\$10,000.00)	\$32,390.00	\$32,390.00
ROE	-0.33%	0.81%	1.12%
Without Credit Loss			
EBIT	\$125,000	\$90,000	\$76,000
Interest	\$85,000.00	\$34,000.00	\$34,000.00
Tax	\$8,400.00	\$11,760.00	\$8,820.00
NI	\$31,600.00	\$44,240.00	\$33,180.00
ROE	1.05%	1.11%	1.14%

考察: Finance: Financial Statement Analysis 重要程度: [■■■□]
 科目: Quantitative Analysis

18. The three lines of defense model (3LoD) has existed in some variation for over 30 years, although it achieved more visibility in 2013 when the Institute of Internal Auditors (IIA) published its initial position paper on the 3LoD, which it updated in 2021 (The IIA’s Three Lines Model, an update of the Three Lines of Defense). GARP recommends firms refer to this update because the IIA “has refined the model to reflect the evolution of risk management within organizations and aim to foster closer collaboration between business functions including internal audit.”

In regard to the 3LoD model, each of the following is true EXCEPT which is false?

- A. The first line defines the organization’s risk appetite but should not attempt to directly influence its risk culture
- B. The first line is responsible for implementing and maintaining effective controls to manage material operational risks
- C. The second line is part of management (and owns the ORM methodology) but provides support to the first line in order to help the firm create and protect value
- D. The third line has the clearest boundaries due to its independence

答案:A
 解析:

A 选项正确。A. False, both clauses. Instead, the following is true: The risk appetite (i.e., risk appetite statement) is developed by senior management and approved by the board. The risk culture is not the sole responsibility but is foundational to the firm’s actual ORMF, such that all employees bear some responsibility for culture.

In regard to (B), (C), and (D), each is TRUE.

考察: Operational Risk 重要程度: [■■■□]
 科目: Quantitative Analysis

19. Liquidity Early Warning Indicators (EWI) can be compared to an automobile’s dashboard signal system where the appearance of a red light points our attention to something that may become a problem if not addressed. Each of the following is a credible red indicator (aka, red flag) in a Liquidity Early Warning Indicator (aka, Liquidity EWI) dashboard EXCEPT which is not a red (red flag) liquidity indicator?

- A. Narrowing debt/CDS spreads
- B. Sudden increase in debt costs
- C. Sudden growth in assets accompanied by volatile liabilities
- D. Rapid decline in the weighted average maturity of liabilities

答案:A

解析:

A 选项正确。 A. FALSE. Instead, the liquidity EWI is: a widening debt/CDS spreads (consistent with a sudden increase in debt costs)

In regard to (B), (C), and (D), each is a candidate Liquidity EWI, per the Study Note: “The Basel Committee on Banking Supervision (BCBS) recommends certain measures as EWI. These measures include [collapsed here, new emphasis ours]:

- An unusual growth in assets, particularly when accompanied by volatile liabilities;
- Debt (credit) spreads *widen*, and/or credit default swap (CDS) spreads *widen*.
- Declining diversity in the makeup of assets and liabilities; Growing currency mismatches;
- When the weighted average of liabilities’ maturity declines;
- Positions going beyond or getting close to regulatory limits;
- Certain product line experience negative trends; The financial condition of the bank weakens; Public press that is negative;
- A downgrade in the credit rating; A decline in the stock price;
- Debt costs increase; Retail and/or wholesale funding costs increase;
- Counterparties becoming nervous about the financial condition of the bank;
- Credit lines are lowered;
- Outflows of retail deposits at an increased pace; Certificates of deposit (CDs) are increasingly redeemed;
- Longer-term funding opportunities become more difficult; and Placing short-term liabilities becomes more difficult”

考察: Liquidity Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

20. In factor theory, an asset’s risk premium (aka, expected excess return) is given by $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ where the asset’s beta is with respect to the stochastic discount factor (SDF; aka, pricing kernel) and (m) in an index of bad time. Let’s make the following assumptions about an asset:

- The riskfree rate is 3.0% such that $E(m) = 1/1.03 = 0.9709$
- The variance(m) equals $30.0\%^2 = 0.090$
- The covariance(r_i, m) equals -0.10680

Which is **nearest** to the asset’s risk premium which refers to the asset’s expected excess return, $E(r_i) - R_f$?

- A. -5.0%
- B. +4.0%
- C. +8.0%
- D. +11.0%

答案:D

解析:

D 选项正确。 The expected excess return, $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ where $\beta(i, m) = \text{cov}(r_i, m) / \text{var}(m)$ and $\lambda(m) = -\text{var}(m) / E(m)$.

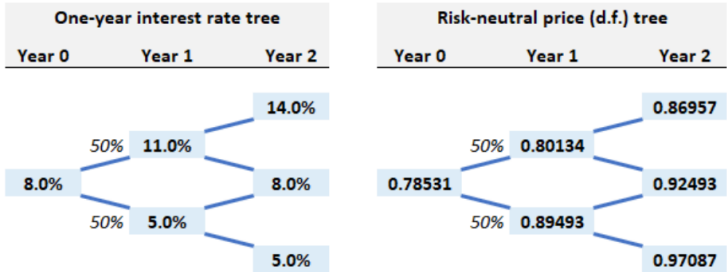
In this case, $\beta(i, m) = \text{cov}(r_i, m) / \text{var}(m) = -0.10680 / 0.090 = -1.186667$, and $\lambda(m) = -\text{var}(m) / E(m) = -0.090 / 0.9709 = -0.0926975$.

Therefore, $E(r_i) - R_f = \beta(i, m) \times \lambda(m) = -1.186667 \times -0.0926975 = 0.1100011$ or about 11.0%.

考察: Finance: Factor Theory 重要程度: [■■■]

科目: Quantitative Analysis

21. Below (in the left-hand panel) is an interest rate tree for the one-year rate where the initial rate of 8.0% jumps 300 basis points to either 11.0% or 5.0% next year. In two years, the one-year rate will be either 2.0%, 8.0%, or 14.0%. Next, we will assume a three-year zero-coupon bond. Without a risk premium, for example, the date 0, state 2 price would be $0.877193 = 1 / (1 + 14\%)$. However, we will instead assume a risk premium. Specifically, investors require compensation of 50 basis points for each year of duration risk. The right-hand panel displays the associated risk-neutral (i.e., , assumes the risk premium) price tree for this three-year zero-coupon bond



In regard to the three-year zero-coupon bond, which of the following is true?

- A. The expected return over the next year is 7.50%
- B. The expected return over the next year is 9.00%
- C. The implied 3-year spot rate term structure is downward sloping (aka, inverted)
- D. Jensen’s inequality implies that expected excess return over two years is less than 50 basis points

答案:B

解析:

B 选项正确。 True: The expected return over the next year is 9.00%.

We don’ t require the formula. The risk premium is 50 basis points for each year of duration risk, and the three-year bond creates two years of duration risk. The expected return, therefore, is 100 basis points above 8.0%.

考察: Fixed Income 重要程度: [■■■]

科目: Quantitative Analysis

22. Old Crow Bank had a credit committee meeting yesterday, and the following is a copy of the notes taken during the meeting:

- All members are present Jan (Chief Risk Officer), Eddie (Head of Compliance), Alyssa (Head of Legal), Don (Head of Tax), and Ashton (Chief Financial Officer).
- Jan initiated the meeting by confirming if everyone had received an email containing approval documents from credit officer Doug, who had originated the loan. Everyone confirmed receiving the email with the documents yesterday.
- Jan provided an overview of the loan for approximately five minutes, after which she proceeded to inquire about the opinions of those present. It was the consensus of all individuals, except for Alyssa, that the loan be approved. Alyssa expressed concern regarding the trustworthiness of the counterparty.

- Jan respects Alyssa’s opinion but disagrees and initiates a vote. Two other committee members agree with Jan, resulting in loan approval.

What is the most probable issue with the credit committee’s meeting?

- A. The approval documents should have been distributed earlier.
- B. Jan monopolized the conversation and did not properly facilitate discussion.
- C. The meeting was not properly documented.
- D. The vote was not unanimous, so the loan should have been rejected.

答案:C

解析:

Option C is correct. In this scenario, all the approval documents were appropriately distributed to the committee before the meeting. Jan did spend time explaining the investment to the committee, but she also helped facilitate discussion among the committee members and listened to everyone’s opinion. Votes do not have to be unanimous, but they should be held unless the decision is unanimous. However, meetings should be accurately documented in case an investment goes awry. In the case of a bad transaction, auditors would want to know what was discussed in the meeting and if the reasoning is behind the investment decision.

考察: Credit Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

23. A bank is using the VaR and stressed VaR (SVaR) market risk framework according to the Basel 2.5 guidelines. The bank’s internal models for market risk have generated the following risk measures for the current trading book positions (VaR and SVaR are in USD millions):

Multiplicative factors, m and m(S)				3.0
Confidence Level	Previous day		60-day average	
	VaR	SVaR	VaR	SVaR
	(mm)	(mm)	(mm)	(mm)
95.00%	\$110.0	\$190.0	\$127.0	\$238.0
99.00%	185.0	360.0	140.0	270.0
99.90%	283.0	740.0	228.0	490.0

The supervisor has set both multiplication factors, m and m(S), to 3.0. What is the correct capital requirement for general market risk for the bank under Basel 2.5? (note: this question inspired by Question 42 of GARP’s 2020 Part 2 Practice Exam).

- A. \$630.0 million
- B. \$810.0 million
- C. \$1,230.0 million
- D. \$2,154.0 million

答案:C

解析:

C 选项正确。 True: \$1,230.0 million

The Basel 2.5 market risk capital requirement requires a 99.0% confidence level and is given by:

$$MRC = \max[VaR(t - 1), m \times VaR(60\text{-day avg})] + \max[SVaR(t - 1), m(S) \times SVaR(60\text{-day avg})]$$

In this case,

$$MRC = \max[185.0, 3.0 \times 140.0] + \max[360.0, 3.0 \times 270.0] = \$1,230.0 \text{ million.}$$

考察: Market Risk 重要程度: [■■■]

科目: Quantitative Analysis

24. If a corporation’s marginal tax rate is 30.0% and it purchases an A-rated municipal bond that offers a 4.20% gross yield (aka, yield to maturity), what is the tax equivalent yield (TEY)?

- A. 2.45%
- B. 3.50%
- C. 6.00%
- D. 16.67%

答案:C

解析:

C 选项正确。 TRUE: 6.00%. $TEY = \text{after-tax return on tax-exempt investment} / (1 - \text{marginal tax rate})$; in this case, $TEY = 4.20\% / (1 - 0.30) = 6.00\%$

考察: Finance: Fixed Income 重要程度: [■■■■]

科目: Foundations of Risk Management

25. Andrew Ang introduces factors and divides them primarily into two types: macro, fundamental-based factors versus investment-style factors. He writes, “Factors drive risk premiums. One set of factors describes fundamental, economywide variables like growth, inflation, volatility, productivity, and demographic risk. Another set consists of tradeable investment styles like the market portfolio, value-growth investing, and momentum investing.” (†)

Ang also claims that the three most important macro factors are growth, inflation, and volatility. His evaluation of these macro factors is based on a long-term historical sample (specifically, 1952:Q1 to 2011:Q4). It is important to qualify the sample window because we cannot be sure that past is prologue; e.g., interest rates experienced two long-term secular trends during this window.

In regard to his historical analysis, each of the following statements is true EXCEPT which is false?

- A. Government and investment-grade bonds performed BETTER during economic recessions than during expansions
- B. Both large and small (cap) stocks perform significantly BETTER during economic expansions than during recessions
- C. During periods of high inflation, all five asset classes perform significantly BETTER than during periods of low inflation
- D. All five assets classes are much MORE VOLATILE during recessions (or periods of low GDP growth) than during expansions

答案:C

解析:

C 选项正确。 False. To be true, this should instead read: During periods of high inflation, all five asset classes perform significantly WORSE than during periods of low inflation.

In regard to (A), (B) and (D), each is TRUE.

考察: Factor Investing 重要程度: [■■■]

科目: Financial Markets and Products

26. Below is the price tree for a stylized constant-maturity Treasury (CMT) swap struck at 2.0% with a notional of \$1.0 million. The risk-neutral probabilities are displayed, specifically: 80.09% (d = 19.91%) from date 0 to date 1; and 65.20% (d = 34.80%) from date 1 to date 2. Using these risk-neutral probabilities returns an expected discounted value (EDV) of \$3,702.11 for the value of the CMT swap. Please notice this exactly matches Tuckman’s Figure 7.8 in his 4th edition value. This is the so-called model value.

Notional	\$1,000,000	
Strike rate	2.00%	
i-rate evolution	T= 0.5	T= 1.0
		3.00%
	2.50%	2.00%
2.0%	1.50%	1.00%
Risk-neutral	80.09%	65.20%
Probabilities:	19.91%	34.80%
Price tree: 2 dates	T= 1.0	
		\$5,000.00
	\$5,719.52	\$0.00
\$3,702.11	-\$4,227.29	-\$5,000.00
\$3,694.81	← Observed Market Price	
0.25%	← OAS	

Assume the observed market price of this CMT swap is \$3,694.81 (as highlighted in purple), which is \$7.30 less than the market price. This implies an option-adjusted spread (OAS) of 25 basis points. Which of the following is the **best** interpretation of this OAS?

- A. The model returns the market price if we adjust the risk-neutral probabilities to 55.09% and 44.91%
- B. The model returns the market price if we assume adjusted cash flows; e.g., \$3,750 instead of \$5,000 on (state 2, date 2)
- C. The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)
- D. We cannot interpret the OAS here because this is a CMT swap, and the OAS requires an embedded option; e.g., a bond with an embedded call option

答案:C

解析:

C 选项正确。 True: The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)

考察: Fixed Income Analysis 重要程度: [■■■]

科目: Quantitative Analysis

27. Strongcore Financial Corporation has purchased a long-term at-the-money (ATM) call option from a risky counterparty. The option's stock and strike price are \$25.00 million. The underlying asset's volatility, σ , is estimated to be 20.0% while the risk-free rate is 3.0% per annum. The European-style option matures in 3.0 years. In regard to the standard normal cumulative distribution function, in this scenario $N(d1)$ equals 0.66750 and $N(d2)$ equals 0.53450.

With respect to its counterparty risk, Strongcore assigns a hazard rate estimate of 6.0% and believes its counterparty could default at any time during the three-year option period; i.e., the cumulative default probability is relevant for this approximation. Further, if its counterparty defaults, Strongcore assigns a conservative recovery rate estimate of 30.0%. In summary, the assumptions are: $S = K = \$25.0$ million, $\sigma = 20.0\%$, $R_f = 3.0\%$, $T = 3.0$ years; λ (i.e., hazard rate) = 6.0%, and $LGD = 70.0\%$.

In this situation, Strongcore does NOT want to compute the standard credit value adjustment (CVA) formula which is the summation of discounted expected exposures, $EE(t)$, over multiple future periods. Instead, Strongcore wants to rely on the simpler “semi-analytical method” that produces an estimate as a function of the position’s current value, the cumulative default probability, $PD(0,T)$, and the loss given default, LGD . Under this simpler approximation, which of the following is **nearest** to the CVA charge for the long option position?

- A. \$15,600
- B. \$90,000
- C. \$516,000
- D. \$1.1 million

答案:C
解析:
C 选项正确。 \$516,000

$CVA = -LGD \times PD(0,T) \times V(t)$. In this case, $V(t) = S \times N(d1) - K \times \exp(-rt) \times N(d2) = 25 \times 0.66750 - 25 \times \exp(-0.030 \times 3) \times 0.53450 = \4.475 million. The cumulative $PD = 1 - \exp(-0.060 \times 3) = 16.473\%$. Therefore the CVA approximation is given by $70\% \times 16.473\% \times 4.475 \text{ million} = \0.5160 million , or \$516,002.

考察: Credit Risk 重要程度:
科目: Quantitative Analysis

28. Andrew is preparing to make a presentation to the board. The presentation’s goal is to introduce a new, proposed operational risk management (ORM) framework and get the board’s feedback on the way to eventual approval. It happens to be the case that his company is publicly traded with a high cost of capital due to its above-average beta. The board members are generally more familiar with market and credit risk but less familiar with operational risk. Therefore, his presentation will review several of the features (and nature) of operational risk, including its dynamic evolution. His presentation will briefly explore the skewed, heavy-tailed feature of operational risk. For candidate frequency distributions, he will illustrate the Poisson and negative binomial. For candidate severity distributions, he will illustrate the log-normal and generalized Pareto Distribution (GPD).

In addition to these features (i.e., dynamic, heavy-tailed), which of the following is TRUE about operational risk?

- A. The easiest way for a high-beta firm to lower its beta is to centralize the operational risk management (ORM) function
- B. Operational risk is highly homogeneous because so-called boundary events define clusters that contain loss events and their shared (aka, common) features
- C. ORM is valuable because it reduces regulatory capital, and regulatory capital is a good indicator of the firm’s quality of risk management
- D. Most causes of operational risk trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes

答案:D
解析:

Option D is correct. Most operational risk causes trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes
In regard to (A), (B), and (C), each is FALSE.

- Operational risk is predominantly idiosyncratic and diffuses within the firm. Beta is a systematic risk.

- Operational risk is highly heterogeneous but interconnected. Further, boundary events are “risk events that materialize in a different category than they cause” -Source: GARP Introduction to Operational Risk and Resilience, Section 1.5)
- According to GARP, “regulatory operational risk capital is not sensitive enough to a firm’s risk profile and thus brings no real incentive to optimize the management of operational risks. In fact, regulatory capital is often a poor indicator of risk management quality.” -Source: Chapter (GARP Introduction to Operational Risk and Resilience, Section 1.7)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

29. Over the next 24 hours, Greenlux State Bank estimates that the following cash inflows and outflows (all figures in millions) will occur:

Deposit withdrawals	\$70.0
Deposit inflows	\$100.0
Scheduled loan repayments	\$60.0
Acceptable loan requests	\$90.0
Borrowings from the money market	\$80.0
Sales of bank assets	\$30.0
Stockholder dividend payments	\$20.0
Revenues from sale of nondeposit services	\$10.0
Repayment of bank borrowings	\$50.0
Operating expenses	\$40.0

What is the bank’s projected net liquidity position?

- A. -30.0 million
- B. +10.0 million
- C. +40.0 million
- D. +90.0 million

答案:B

解析:

B. TRUE: +10.0 million

Inflows include: Deposit inflows = 100+ Scheduled loan repayments = 60+ Borrowings from the money market = 80+ Sales of bank assets = 30+ Revenues from sale of nondeposit services = 10; for total inflows of \$100 + \$60 + \$80 + \$30 + \$10 = 280.0 million.

Outflows include: Deposit withdrawals = 70+ Acceptable loan requests = 90+ Stockholder dividend payments = 20+ Re- payment of bank borrowings = 50+ Operating expenses = 40; for total outflows of \$70 + \$90 + \$20 + \$50 + \$40 = \$270.0 million

Therefore, the projected net liquidity position = \$280 – 270 = +10.0 million.

考察: Finance 重要程度: [■ ■ □]

科目: Quantitative Analysis

30. Emily manages a \$20.0 million portfolio allocated into two positions (aka, two components): a Technology ETF, and a Real Estate Investment Trust (REIT) ETF. She compares four different allocations, where the weight assigned to the Tech ETF component is either 35.0%, 50.0%, 65%, or 80.0%.

	Mix (Allocation) Scenarios			
	#1	#2	#3	#4
Weight in Tech ETF (%)	35.0%	50.0%	65.0%	80.0%
Position (000s)	\$7,000	\$10,000	\$13,000	\$16,000
Individual VaR (000s)	\$2,303	\$3,290	\$4,277	\$5,264
Marginal VaR	0.020	0.118	0.231	0.305

	#1	#2	#3	#4
	#1	#2	#3	#4
Weight in REIT ETF (%)	65.0%	50.0%	35.0%	20.0%
Position, USD (000s)	\$13,000	\$10,000	\$7,000	\$4,000
Individual VaR (000s)	\$6,415	\$4,935	\$3,454	\$1,974
Marginal VaR	0.461	0.386	0.232	0.039

Undiversified Portfolio VaR	\$8,718	\$8,224	\$7,731	\$7,237
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If her goal is to minimize the portfolio’ s diversified value at risk (VaR), which of the above allocations is **best**?

- A. Mix #1 with 35.0% allocated to the Tech ETF
- B. Mix #2 with 50.0% allocated to the Tech ETF
- C. Mix #3 with 65.0% allocated to the Tech ETF
- D. Mix #4 with 80.0% allocated to the Tech ETF

答案:C

解析:

Option C is correct. Mix #3 with 65.0% allocated to the Tech ETF.

Risk is minimized (i.e., the portfolio’ s diversified VaR is minimized) when the marginal VaRs are constant. Among the four mixes, mix #3 is clearly the choice where the marginal VaRs are nearest (i.e., 0.231 versus 0.232 is almost an exact match). Mix #1 has the greatest diversified VaR. Starting from Mix #1, risk is reduced by ADDING to the component with the lower marginal VaR (in this mix, the marginal VaR of the Tech ETF is only 0.020 while the marginal VaR of the REIT ETF is 0.461) and REDUCING the component with the higher marginal VaR. If we thusly start with Mix #1 and add 15.0% to the Tech ETF, then we “arrive” at Mix #2. However, in Mix #2 the Marginal VaR of the REIT ETF is still much higher than the Marginal VaR of the Tech ETF (i.e., 0.386 versus 0.118) so we can continue to shift toward the position with the lower Marginal VaR (which is still the Tech ETF). If we add another 15% to the Tech ETF, then we arrive at Mix #3: here the Marginal VaRs are almost equal, so we are very close to the allocation that minimizes the portfolio’ s diversified VaR. For the record, the diversified portfolio VaR for the four mixes are \$6,131.14 (Mix #1), \$5,043.03 (Mix #2), \$4,621.42 (Mix #3), and \$5,036.58 (Mix #4).

考察: Finance: Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

31. In comparison to a yield-based DV01 hedge, which of the following is the **best** argument in favor of a regression hedge?

- A. The regression hedge minimizes the profit and loss (P&L) variance of the hedged position
- B. The regression hedge is based on economic theory and future developments rather than empirical analysis
- C. The regression hedge does not require any out-of-sample test(s), which are often unreliable if the sample is small
- D. The regression hedge solves for a risk-weight which qualifies as a valid risk-weighted asset (RWA) under Basel regulations

答案:A

解析:

A 选项正确。 True: The regression hedge minimizes the profit and loss (P&L) variance of the hedged position.

As Tuckman explains (emphasis ours), “The **best argument for the regression hedge** is actually not the earlier assumption that bond yields change exactly according to the regression model. Write the P&L of the position as ... formula 6.11 ... where

the negative signs reflect that a positive face amount with a positive change (i.e., increase) in yield lowers P&L. It can then be shown that the regression hedge in Equation (6.8) minimizes the variance. In other words, **to the extent that P&L variance is an appropriate measure of risk, the regression hedge minimizes the risk of the hedged position.**” –Tuckman, Bruce; Serrat, Angel. Fixed Income Securities (Wiley Finance) (p. 159). Wiley. Kindle Edition.

In regard to (B), (C), and (D), each is FALSE. Instead, the following are true statements:

- The regression hedge is based on empirical analysis but not economic theory and future developments
- The regression hedge typically does require (or at least prefer) out-of-sample test(s)
- The risk weight is the beta coefficient and has nothing to do with risk-weighted asset (RWA) under Basel regulations

考察: Fixed Income: Hedging 重要程度: [■ ■ □]

科目: Quantitative Analysis

32. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability

答案:D

解析:

D 选项正确。 Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability

Wrong-way (credit) risk is adverse correlation between the exposure and the credit quality, so wrong-way risk is a positive (negative) correlation between exposure and default probability (credit quality). This is wrong-way risk because greater exposure is associated with a lower likelihood of payment. Algebraically, wrong-way (or right-way) risk is some dependence between exposure/EAD and PD. We normally write $EL\% = PD \times EAD$ but that assumes independence between PD and EAD, so a way to think about WWR is simply when $EL > PD \times EAD$, because $EL \neq PD \times EAD$ is also a condition for non-independence.

In regard to (A), (B) and (C), each is an example of RIGHT-WAY risk.

考察: Credit Risk: Wrong-Way Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

33. According to the FDIC, model risk management starts with robust model development, implementation, and use (aka, the first element. The second element is a sound validation process. The third element is governance). Which of the following is a TRUE statement about this *first element* (development, implementation and use) of model risk management?

- A. Model development should be a straightforward and routine technical process
- B. To exercise conservatism, pick an extreme point on a given modeled distribution
- C. Banks should never model output with judgmental or qualitative adjustments
- D. Although the nature of testing varies by the context and type of model, testing is an integral part of model development

答案:D

解析:

D. TRUE: Although the nature of testing varies by the context and type of model, testing is an integral part of model development

In regard to (A), (B) and (C), each is FALSE.

- Model development is neither straightforward nor a routine technical process!
- To pick an extreme point on a given modeled distribution is not necessarily conservative if the distribution is mis-specified or mis-estimated in the first place.
- Judgmental or qualitative adjustments are a KEY part of model development/implementation/use: “In addition, a considerable amount of subjective judgment is exercised at various stages of model development, implementation, use, and validation. It is important for decision makers to recognize that this subjectivity elevates the importance of sound and comprehensive model risk management processes.” ... “Banks should ensure that the development of the more judgmental and qualitative aspects of their models is also sound. In some cases, banks may take statistical output from a model and modify it with judgmental or qualitative adjustments as part of model development.”

考察: Operational Risk: Model Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

34. Suppose that Acme Bank reports interest-sensitive assets (ISA) of \$490.0 million and interest-sensitive liabilities (ISL) of \$610.0 million. Recall that the relative interest-sensitive gap (aka, Relative IS GAP) is equal to the Dollar IS GAP divided (aka, scaled) by the size of the bank (where IS assets is a valid measure of size). Respectively, what is the bank’s relative IS GAP; its interest-sensitivity ratio (ISR); and the impact of rising interest rates on its net interest margin (NIM)?

- A. The Relative IS GAP is -120.0; the ISR is 1.245; rising interest rates will lower the NIM
- B. The Relative IS GAP is -0.245; the ISR is 0.803; rising interest rates will lower the NIM
- C. The Relative IS GAP is -0.245; the ISR is 1.245; rising interest rates will increase the NIM
- D. The Relative IS GAP is +4.08; the ISR is 0.803; rising interest rates will increase the NIM

答案:B

解析:

B 选项正确。 The Dollar IS GAP = ISA - ISL = \$490.0 - \$610.0 = -120.0. The Relative IS GAP = $-120.0/490.0 = -0.245$. The ISR = $490.0/610.0 = 0.803$. This bank is liability-sensitive such that rising interest rates will lower its net interest margin (NIM).

考察: Market Risk: Interest Rate Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

35. You work for the Private Client division of an international bank. The bank’s Private Client Advisors work with high net worth (NNW) individuals and families. Your boss has tasked you to perform due diligence on a roster of hedge funds; the funds that survive your due diligence will become candidates for presentation (and recommendation) to suitable customers of the bank who will become investors in the fund(s).

Your boss says that your due diligence should include a careful screen for fraud risk on behalf of the investors. In the case of your due diligence, which of the following is the most important ingredient?

- A. Diligent application of a comprehensive checklist including a list of specific questions
- B. Strong pre-existing personal relationships among the fund’s managers and all related parties
- C. Guarantees by the fund of exceptional performance (e.g., > 30% per annum) with little or no risk
- D. The investors’ membership in an affinity group, preferably a religious group due to high correlation with ethics

答案:A

解析:

A. TRUE: Diligent application of a comprehensive checklist including a list of specific questions

The authors recommend reviewing the SEC website including its list of questions; e.g., Red Flags of Investment Fraud Checklist at <https://www.investor.gov/>.

In regard to (B), (C) and (D), each is false:

- Strong pre-existing personal relationships with related parties is the **opposite of the independence** that investors should require between fund managers and related parties.
- Guarantees of exceptional performance with little or no risk is a classic **red flag**.
- In regard to affinity groups: these are notorious targets of fraud because participants tend to be trusting of their peers.

考察: Finance: Due Diligence 重要程度:

科目: Valuation and Risk Models

36. Which of the following is the BEST summary of the purpose of a copula function?

- A. To utilize the Gaussian correlation matrix such that tractable pairwise correlation parameters replace intractable dependence values
- B. To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure
- C. To translate a vector of outcomes into quantiles and map the quantiles to probabilities such that virtually any combination of multiple outcomes can be expressed as a single vector
- D. To decompose a posterior conditional distribution into a combination of the prior probability, the likelihood, and the marginal probability, typically by identifying Cholesky’s decomposition

答案:B

解析:

B 选项正确。 To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure

考察: Quantitative Analysis: Copulas 重要程度: [■□□□□]

科目: Foundations of Risk Management

37. Retail exposures include the following: credit cards, installment loans (e.g., personal finance, educational loans, auto loans, leasing), revolving credits (e.g., overdrafts, home equity lines of credit), and residential mortgages. Each of the following tends to be a stronger feature of retail credit risk rather than corporate credit risk EXCEPT which is more typical of corporate credit risk?

- A. Portfolio tends to behave like a well-diversified portfolio in normal markets
- B. Default by a single customer is never expensive enough to threaten a bank
- C. A rise in defaults is often signaled in advance by a change in customer behavior
- D. Portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures

答案:D

解析:

D 选项正确。 This tends to be a feature of CORPORATE credit risk: portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures.

- In regard to (A), (B) and (C), each is a tendency of retail credit risk.

考察: Credit Risk: Retail vs Corporate 重要程度: [■■□□]

科目: Quantitative Analysis

38. According to the Basel Committee on Banking Supervision (BCBS, 2018), in regard to cyber-security and cyber-resilience metrics, “Some jurisdictions have methodologies to assess or benchmark regulated institutions’ cyber-security and resilience ...

None of these methodologies produce quantitative metrics or risk indicators comparable to those available for financial risks and resilience; e.g., standardized quantitative metrics where established data are available. Instead, indicators provide information on regulated institutions’ approach to building and ensuring cyber-security and resilience more broadly. Supervisory authorities also rely on entities’ own management information, although this differs across entities and is not yet mature.” (†)

With respect to current cybersecurity and resilience metrics, each of the following statements is true EXCEPT which is inaccurate?

- A. No single cybersecurity/resilience metric in isolation is sufficient
- B. Patch aging (aka, days to patch) is a widespread and comparable metric
- C. Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt
- D. The penetration ratio (i.e., percentage of banks in a jurisdiction that can be penetrated) is a useful metric that compares a supervisor’s effectiveness to other jurisdictions

答案:D

解析:

D 选项正确。 False. The displayed penetration ratio is nonsensical (also to compare supervisor effectiveness), but penetration tests (penetration testing) are discussed several times by BCBS as an effective cyber-risk control approach. However, please do note that penetration testing (by type: count and finding rating) is a common and effective cyber-resilience metric. See 4.2.2. Penetration testing.

In regard to (A), (B) and (C), each is TRUE. Specifically:

- No single cybersecurity/resilience metric in isolation is sufficient: “Collectively, these indicators [i.e., cyber-metrics collated by regulated entities] can inform on the broad adequacy of an institution’ s cyber- and operational resilience levels for its business needs and risk appetite. However, no single item taken in isolation is seen as a sufficient metric, and no standard set of indicators has been identified so far to provide a meaningful benchmark.” (†)
- Patch aging (aka, days to patch) is a widespread and comparable metric: “Annex C contains cyber-centric metrics collated by a sample set of regulated institutions for decision-making bodies (boards and board sub-committees). It is notable that the data provided typically allow for trend information so that the reviewer can assess if the situation is getting better or worse. Some metrics track compliance with internal policies while others measure inherent risk. Patch ageing in particular is a widespread and comparable metric.” (†)
- Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt: “Backward-looking indicators comment on past performance as an indicator of future performance, which is reasonable when institutions’ operations and risk environment are relatively stable over time and more or less independent from outside influences. However, cyber-risk frustrates this because adversaries are dynamic, themselves adapting to institutions’ responses and protective measures, sometimes changing their tactics and strategies even in the space of a single cyber-incident. Distributed denial of service (DDOS) incidents are a good example, where the volume and scale of disrupted internet traffic generated has increased significantly in the last two years and adversaries adapt their techniques in response to an institution’ s defences. While backward-looking metrics continue to be important, jurisdictions are increasingly recognising the need for forward-looking indicators as direct and indirect metrics of resilience, indicating whether a regulated institution is likely to be more or less resilient in the event of a risk crystallising.” (†)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

39. Venkat explains that “all risk management frameworks start with a governance structure that defines the roles and responsibilities of various bank employees and committees in overseeing risk-related activities,” and effective governance includes the oversight of intraday liquidity risk. In regard to the governance structure of intraday liquidity risk management which of the following statements is TRUE?

- A. Intraday liquidity risk should be accepted as a cost of doing business

- B. Roles and responsibilities should be defined within the eight lines (i.e., two times four) of defense model
- C. PCS Systems should only be a source of funds; if PCS becomes a use of funds, then a yellow flag should be triggered
- D. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.

答案:D

解析:

Option D is correct. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.

In regard to (A), (B) and (C), each is FALSE.

- Venkat(†) explains that Active Risk Management is characteristic of the governance of intraday LRM: “Active risk management. In many institutions, intraday liquidity risk is accepted as a cost of doing business and is not as actively managed with the same level of rigor as other types of enterprise risk or even other liquidity risks. The leading banks with large volumes of PCS activities recognize the criticality of understanding and working to reduce their intraday liquidity risks. These institutions classify settlement and systemic risks as components of their risk taxonomy, and critically, incorporate them into the firm’ s risk appetite framework” (†)
- Integration with risk governance includes the THREE lines of defense model: Treasury, Corporate Risk Management, and Internal Audit. Says Venkat(†), “The intraday liquidity risk management framework follows the industry’ s three lines of defense model, with particular emphasis on expertise in the second line of defense to coordinate across the institution ... Corporate Risk Management is the second line of defense, responsible for overseeing funding-related policy and procedures: advising in the development of risk management programs, monitoring the ongoing risk-taking activities across all of a bank’ s funding desks, aggregating reporting across the bank, and providing an independent view of the effectiveness of the bank’ s overall intraday liquidity risk management programs.” (†)
- Payment, clearing, and settlement (PCS) Systems can be either a source or use of funds: “Settlements at PCS systems. Most PCS systems have one settlement per day, with many occurring in the late afternoon timeframe. This may serve as either a source or use of funds depending on the net position of a participant on any given day.” (†)

考察: Liquidity Risk Management 重要程度: ☒☒☐

科目: Valuation and Risk Models

40. Hedge funds are less regulated than mutual funds. According to the authors (Fung and Hsieh), this lighter regulation of hedge funds offers genuine benefits but also creates unintended consequences. Which of the following is both a negative consequence and a TRUE statement about this relatively light regulation?

- A. Lack of reliable, standardized, and long-term performance reporting
- B. Lack of agreement on the specific definitions of absolute return and alpha
- C. Ability of global macro funds to secretly drift into various strategies and styles with reckless impunity
- D. Ability of hedge funds to advertise (and specifically solicit) to individual investors without any restrictions

答案:A

解析:

A 选项正确。 Lack of reliable, long-term performance reporting

According to the authors (emphasis ours): “However, being lightly regulated does have unintended consequences. For one, **performance records of hedge funds are generally not standardized and available reports are prone to measurement errors.** Until the early 1990s, the historical performance of hedge funds was often as private as their investors, and the assessment of hedge fund performance was as much art as science ... the **events of 1994** prompted investors to re-examine the perceived diversification of their hedge fund holdings. In turn, this aided the development of electronically available hedge fund databases with which more formal analysis of hedge fund performance and the attendant risks can be conducted. It was the arrival of electronic databases that made academic research in hedge funds a feasible proposition.” —Source: Chapter 17 of Handbook of the Economics of Finance, Volume 2B, edited by G. Constantinides, M. Harris, and R. Stulz.

- In regard to (B) and (D), these are false statements.
- In regard to (C), it is a feature (aka, advantage) of global macro funds that they can utilize various styles opportunistically.

考察: Finance: Hedge Funds 重要程度: [■ ■ □]

科目: Quantitative Analysis